



Fault hydromechanics control the emergence of seismic and aseismic slip patterns

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ABSTRACT

Understanding why some fault systems produce seismic swarms while others evolve into foreshock–mainshock–aftershock sequences remains a central challenge in seismology. Fluids are known to play a key role, but numerical models are still far from reproducing the full complexity of solid–fluid interactions and the entire range of fault slip behavior. Here, we use a fully coupled 2D poro-visco-elasto-plastic fault model with rate-and-state friction to explore how pore-fluid pressure and permeability structure jointly shape fault slip behavior. The model resolves off-fault poroelastic deformation and fluid flow, and incorporates a slip- and time-dependent permeability evolution law. By systematically varying the along-strike widths of low-permeability barriers and over-pressurized patches relative to the nucleation size, as well as the healing timescale of permeability, we map out a continuum of dynamic regimes — from distributed, similar-sized clustering to system-size ruptures. Low-permeability barriers act as seismic asperities that promote rupture, whereas high pore-fluid pressure patches favor aseismic slip that redistributes stress. Following seismic ruptures that damage and unseal the fault zone, long permeability-healing times allow the resulting high-permeability pathways to remain open over multiple events, enhancing fluid redistribution and favoring swarm-like activity. In contrast, short healing times rapidly reseal these damaged pathways after each event, trapping fluids between barriers, amplifying stress concentrations, and promoting large, system-size dynamic ruptures. These results show how evolving fault-zone hydromechanics can generate diverse seismic sequences from common underlying physics, providing a framework for interpreting the natural variability in fault slip modes.

1. Introduction

Earthquakes are observed to cluster in space and time. Two main groups of earthquake clustering are typical foreshock-mainshock-aftershock sequences (hereafter “FMA sequences”) and seismic swarms, each involving multiple seismic events, but differing significantly in their characteristics (Utsu, 1961). Specifically, FMA sequences feature a primary mainshock, occasionally preceded by smaller foreshocks and typically followed by aftershocks, whose temporal decay rate decreases inversely with time, as described by Omori’s law (Richter, 1958). In contrast, earthquake swarms consist of a series of closely spaced seismic events occurring in a localized area over a period of time, characterized by the absence of a distinct mainshock (Mogi, 1963; Scholz, 2019). The term swarm has been historically used by seismologists to describe temporal clusters of earthquakes that do not conform to Omori’s law (Hainzl and Fischer, 2002). However, a clear and universally accepted definition for earthquake swarms remains elusive, reflecting

the complexity and variability of earthquake clusters that represents more a continuum between FMA and swarm-like sequences rather than distinct classes (Passarelli et al., 2025). Migration velocities in swarm-like sequences can vary significantly, ranging from meters per day to kilometers per hour (Passarelli et al., 2021, 2018; Danré et al., 2022, 2024), and both swarms and FMA sequences can span time periods from weeks to years. The mechanisms governing these behaviors are not yet fully understood, due in part to observational limitations and the intrinsic challenges of detecting and characterizing small-scale fault processes.

Recent advancements in machine learning techniques, such as template-matching detection (Ross et al., 2018) and deep neural network algorithms (Zhu and Beroza, 2019; Mousavi et al., 2020), have significantly enhanced our ability to detect smaller seismic events that were previously missed by conventional methods (Ross et al., 2019). These techniques have resulted in higher resolution catalogs

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of hidden events, and shown and increased complexity of seismic sequences that are crucial for understanding fault dynamics, strain localization, and the mechanisms of earthquake nucleation (Skoumal et al., 2019; Ross et al., 2020; Li and Zhan, 2018; Lee and Douilly, 2023; Essing and Poli, 2022; Liu et al., 2022; Meng et al., 2019). Despite these advancements, linking seismicity to heterogeneous fault structures and dynamic stresses remains a significant challenge. As a result, the mechanisms that trigger both FMA and earthquake swarms are still not well understood.

A range of physical processes have been proposed to influence the character of FMA and earthquake swarms, including magmatic intrusion in volcanic regions (Passarelli et al., 2015; Sigmundsson et al., 2015; Woods et al., 2019; Dreger et al., 2000; Wilding et al., 2023), aseismic creep (Vidale and Shearer, 2006; Lohman and McGuire, 2007), and metamorphic dehydration reactions that release fluids (Hill et al., 1975; Im et al., 2021). Fluids are thought to play a particularly important role in earthquake triggering, as solid–fluid interactions and poroelastic effects can destabilize faults and lead to a spectrum of seismic and aseismic slip behaviors (Dunham and Rice, 2008; Miller, 2013; Jansen et al., 2019; Ross et al., 2017; Miller, 2020; Hatch et al., 2020; Brown et al., 2005; Bürgmann, 2018). A key factor in this variability is the hydraulic structure of faults, which governs how fluids migrate and accumulate. Fault zones can act as both conduits and barriers to fluid flow, producing spatial variations in pore-fluid pressure that influence local fault strength (Bense and Person, 2006; Manga et al., 2012; Lockner et al., 2000; Wibberley and Shimamoto, 2005). Permeability, which controls fluid mobility, can vary significantly depending on fault geometry, lithology, stress, and deformation history (Caine et al., 1996; Bense et al., 2013; Saffer and Tobin, 2011). These variations — and their time-dependent evolution — are believed to play a central role in modulating fault slip modes (Bürgmann, 2018).

Substantial research within the earthquake modeling community has focused on understanding the frictional dependency of seismic and aseismic slip (Chen and Lapusta, 2009; Liu and Rice, 2005; Lapusta et al., 2000; Jiang et al., 2022). This includes complex fault geometry (at larger scales) (Romanet et al., 2018), fault roughness (at smaller scales) (Cattania and Segall, 2021; Tal et al., 2018; Fang and Dunham, 2013; Ozawa et al., 2019), and dynamic weakening processes such as thermal pressurization (Viesca and Garagash, 2015; Rice, 2006; Bizzarri and Cocco, 2006; Perry et al., 2020), thermal effects (Barbot, 2022; Dal Zilio et al., 2022b; Allison and Dunham, 2018), and solid–fluid coupling with poroelastic effects (Segall and Rice, 1995; Dal Zilio et al., 2022a; Cappa et al., 2022; Zhu et al., 2020; Garagash and Germanovich, 2012; Heimissson et al., 2019; Zencher et al., 2006). However, despite these advances, many simulations tend to simplify the complex interactions within fully coupled solid–fluid systems by predominantly focusing on on-fault pore-pressure diffusion in fluid injection models. In these models, fluid pressure influences the effective normal stress, yet the reverse interaction — poroelastic feedback where stress affects fluid flow — is often neglected. While some recent studies have incorporated off-fault fluid exchange (Marguin and Simpson, 2023, 2024a,b) and variations of permeability with slip history (Zhu et al., 2020), the two-way interactions between fault slip, stress, and fluid migration in the entire fault domain remain poorly resolved, potentially limiting our understanding of the full range of conditions under which different slip modes arise.

In this study, we investigate the interplay among fault slip, poroelasticity, and hydro-mechanical processes. To achieve this, we introduce a fully coupled solid–fluid modeling approach that captures the hydro-mechanical response of a compressible solid–fluid system with a poro-visco-elasto-plastic rheology governed by rate- and state-dependent friction. Our simulations capture all stages of the earthquake cycle, from long-term fault loading to short-term seismic and aseismic slips, poroelastic deformation, and fluid flow on and off the fault. In Section 2, we present the description of governing methods and model setup in this work. In Section 3, we first show a set of simulations of

earthquake sequences with time-invariant permeability to isolate the effects of permeability barriers and high pore-fluid pressure patches. We then introduce permeability evolution to investigate how variations in permeability structure and fluid diffusion, particularly the healing timescale of fault permeability, modulate the distribution of pore-fluid pressure and influence rupture style. Finally, Section 4 discusses the comparison between modeling results and observations, and outlines the model limitations. Our models imply that observed variability in seismic sequences may arise from common underlying processes, shaped by evolving interactions between fluid flow and fault strength, rather than fundamentally distinct triggering mechanisms.

2. Methodology

In this study, we use the Hydro-Mechanical-Earthquake Cycles (H-MECs) numerical code (Dal Zilio et al., 2022a), which offers a comprehensive framework for simulating fully coupled solid–fluid fault processes. The code resolves the hydro-mechanical behavior of a compressible solid–fluid system with poro-visco-elasto-plastic rheology, incorporates rate-and-state friction, accounts for full inertia, and employs adaptive time stepping to efficiently capture both seismic and aseismic slip. Further details are provided on the governing equations, the rheology, the rate- and state-friction law, the permeability evolution law, and our model setup.

2.1. Governing equations

The momentum conservation of the fluid and solid governs the fully coupled solid–fluid system (see Appendix A). To reduce computational cost and increase resolution, we first solve the displacement in the z -direction, and then the flow equation for the fluid in-plane motion (x – y plane). The 2-D antiplane motion implies an infinitely long fault along the z -axis, meaning no variance along this axis.

2.2. Rheology and friction law

The rheology of the model zone is described by a visco-elasto-plastic Maxwell model (Dal Zilio et al., 2022a). We assume that strain rate can be decomposed into its elastic, viscous, and plastic components as

$$\dot{\epsilon}'_{ij} = [\dot{\epsilon}'_{ij}]_{\text{viscous}} + [\dot{\epsilon}'_{ij}]_{\text{elastic}} + [\dot{\epsilon}'_{ij}]_{\text{plastic}}. \quad (1)$$

Details of each component are in Appendix B. The yield strength of the fault zone is governed by the rate- and state-friction law. We incorporate the rheology model and the invariant formation of the regularized rate- and state-friction law (Dal Zilio et al., 2022b) as

$$\tau_s = f p_{\text{eff}} = a \operatorname{arcsinh} \left[\frac{V(t)}{2V_0} \exp \left(\frac{f_0 + b \ln \frac{\Theta(t)V_0}{L}}{a} \right) \right] (p^t - p^f). \quad (2)$$

Here, the friction coefficient f relates shear stress τ_s to effective pressure p_{eff} (the difference between total pressure p^t and pore-fluid pressure p^f) on the fault and depends on transient slip rate $V(t)$ and state $\Theta(t)$ that follows the aging law ($\dot{\Theta} = 1 - V\Theta/L$). f_0 is the reference friction coefficient, V_0 is steady-state slip rate, L is the characteristic slip distance. a and b are constitutive parameters of material property (Dieterich, 1979, 1981; Ruina, 1983; Marone, 1998).

2.3. Permeability evolution law

Our model is fully solid–fluid coupled, meaning that deformation of the solid and the pore fluid mutually influence one another. This coupling arises directly from the governing equations (Eqs. (A.1a)–(A.1d)), which link fault deformation, pore-fluid pressure, and fluid flow. Building on this framework, we incorporate a permeability evolution law (Zhu et al., 2020) to capture changes in fault-zone permeability during fault slip and healing. The evolution of permeability is driven by the feedback between fault slip and deformation, which elevates

permeability, and fault healing and sealing processes (such as pressure solution transfer), which reduce permeability (Renard et al., 2000; Gratier et al., 2003; Tenthorey et al., 2003). The change in permeability is represented by $k_{\min} = 10^{-21} \text{ m}^2$ and $k_{\max} = 10^{-15} \text{ m}^2$, which define the minimum and maximum bounds on permeability constrained by laboratory experiments (Lockner et al., 2000; Wibberley, 2002; Mitchell and Faulkner, 2008; Dong et al., 2010; Evans et al., 1997) and in situ measurements (Xue et al., 2013; Osten et al., 2024). This dynamic permeability variation leads to fluctuations in pore-fluid pressure and effective pressure. Permeability evolution within the fault zone in our model is described by the following equations (Zhu et al., 2020):

$$k = k_{\min} + (k^* - k_{\min}) e^{-(\sigma - p)/\sigma^*} \quad (3a)$$

$$\frac{\partial k^*}{\partial t} = -\frac{V}{l_k} (k^* - k_{\max}) - \frac{1}{t_k} (k^* - k_{\min}) \quad (3b)$$

where $\sigma^* = 30 \text{ MPa}$ is the empirically determined stress-sensitivity parameter (Dong et al., 2010; Lockner et al., 2000; Mitchell and Faulkner, 2008), k^* is the reference permeability, l_k is the permeability enhancement evolution distance, and t_k is the characteristic healing time. V is the slip rate on the fault.

Eq. (3a) accounts for the observed decrease in permeability with increasing effective normal stress in laboratory studies (Faulkner and Rutter, 2001; Dong et al., 2010; Evans et al., 1997), and Fig. 1c shows the initial permeability profile implemented with the permeability evolution law as described in Eq. (3a). Eq. (3b) describes how permeability increases with slip during seismic or aseismic events (weakening) and decreases with time during healing processes after slip. The first term on the right-hand side of Eq. (3b) describes the increase in permeability towards the maximum value over the slip distance l_k . The second term represents the permeability drop over the healing time t_k . The weakening process dominates during earthquakes, while the healing process dominates during the interseismic period. We note that the permeability evolution law is a simplified idealization that captures the essential processes of permeability increase during slip and decrease during healing. While it is motivated by laboratory and field observations of permeability changes, it is not directly constrained by observations. l_k and t_k are poorly constrained in nature, owing to the complex processes that increase or decrease permeability. Notably, the recovery of permeability to its pre-seismic level can vary from hours to years (Manga et al., 2012). During this process, little evidence exists regarding the range of l_k . In our study, to investigate the influence of l_k and t_k on fault slip behavior across a broad parameter space, we test t_k from 10 h to 10 years and l_k from 0.1 m to 1 m (Zhu et al., 2020).

2.4. Model geometry and initial conditions

Our starting point is the geological picture in Fig. 1a. Strike-slip faults are not single, uniform planes but damage zones made of lenses and layers with strongly contrasting hydraulic properties (Ross et al., 2020; Ben-Zion and Sammis, 2003; Bense et al., 2013). Low-permeability bands (e.g., clay smears, cemented lenses) act as permeability barriers, whereas more permeable segments and intersections of subsidiary faults form channelized pathways. Fluids sourced from depth can therefore percolate upward, become trapped against barriers, and generate laterally heterogeneous pore-fluid pressure with overpressurized patches embedded in a more normally pressured fault zone.

To study the mechanical consequences of such heterogeneities, we construct a two-dimensional anti-plane shear model that represents a vertical section through the fault zone (Fig. 1b). The domain extends 5 km along dip y , representing a fault segment embedded at depth rather than the shallowest portion of the seismogenic zone. A vertical strike-slip fault is located at $x = L_x/2$ and is embedded in a homogeneous poro-visco-elasto-plastic medium. Along the fault we prescribe a rate-and-state friction law with a central velocity-weakening (VW) segment surrounded by velocity-strengthening (VS) regions (Fig. 1c,

left). The VW segment occupies the central 4.8 km, while the top and bottom 100 m are VS, so that seismic slip can nucleate only in the central portion whereas the boundaries remain stably sliding. These VS regions are introduced primarily to provide stable sliding boundary conditions, rather than to represent the full thickness of shallow or deep stable fault zones. This configuration represents a 5 km segment at depth that is loaded at a constant far-field shear rate $\pm V_p/2$ and is suitable for resolving relatively small and large earthquake sequences. All frictional and bulk parameters are listed in Table 1.

Rather than resolving the long-term percolation of fluids into the fault zone, which acts over thousands of years or longer, we prescribe a *post-percolation* state in which fluids have already migrated and established a heterogeneous pressure field. The corresponding permeability and pore-pressure distributions are designed to represent along-dip heterogeneity within a fault segment at depth, rather than a depth-dependent profile of the crust. Specifically, we impose initial pore-fluid pressure p^f that is lower inside permeability barriers (10 MPa) and higher in the adjacent permeable patches (25 MPa). For a fixed lithostatic total pressure p^t , this corresponds to higher effective pressure $p_{\text{eff}} = p^t - p^f$ in barriers (30 MPa) and lower effective pressure in high-pressure patches (5 MPa). In other words, the imposed permeability architecture is used to represent the mature, heterogeneous effective normal stress expected after long-term fluid circulation in a layered fault zone, as sketched in Fig. 1a. This configuration represents a state in which fluids preferentially accumulate in more permeable regions adjacent to barriers, while reduced hydraulic connectivity within low-permeability barriers limits fluid storage, resulting in relatively lower pore-fluid pressure within the barriers. Porosity is kept constant at 1%, so that variations in permeability dominate the hydraulic response.

Permeability is not fixed during slip. In addition to the fully coupled poromechanical governing equations, we allow k to evolve according to the permeability evolution law of Zhu et al. (2020) (see Section 2.3), which links changes in permeability to the local slip and healing history on the fault. This law permits permeability to increase by up to three orders of magnitude, with upper bounds of 10^{-18} m^2 in the barriers and 10^{-15} m^2 in the initially high-permeability patches, thereby capturing damage-induced permeability enhancement and subsequent relaxation while preserving a strong contrast between barriers and channels. In our numerical experiments, we systematically vary the along-dip widths of permeability barriers (W_B) and high pore-fluid pressure patches (W_P) to quantify how the scale and arrangement of these hydraulic heterogeneities control fluid redistribution, effective pressure, and the resulting patterns of aseismic and seismic slip along the fault.

The initiation of aseismic and seismic events starts from the loading of the left ($x = 0$) and right ($x = L_x$) boundary with velocity V_p in the direction parallel to the fault plane, where $V_p = 2 \times 10^{-9} \text{ m/s}$ corresponds to the long-term loading rate. By applying Dirichlet boundary conditions for $x = 0$ and $x = L_x$, horizontal velocity component v_x and vertical velocity component v_y are zero. Boundary conditions along $y = 0$ and $y = L_y$ requires that normal velocity component is zero and other velocity components do not change across the boundary, which is known as free slip.

As fluids are assumed to percolate within the fault zone, gravity and depth dependence are neglected. The source of fluid is not injection but the initialized pore-fluid pressure distribution in the model. This simplification is consistent with our main objective: to investigate earthquake sequences on a fault profile with a strong permeability heterogeneity, as suggested by seismological observations (Ross et al., 2020).

2.5. Seismic/aseismic threshold

The slip behavior is categorized as either aseismic or seismic, based on the maximum slip rate $V_{\max}$ of an event compared to the threshold velocity (Rubin and Ampuero, 2005) defined as

$$V_{\text{th}} = \frac{2 a p_{\text{eff}} c_s}{\mu} \quad (4)$$

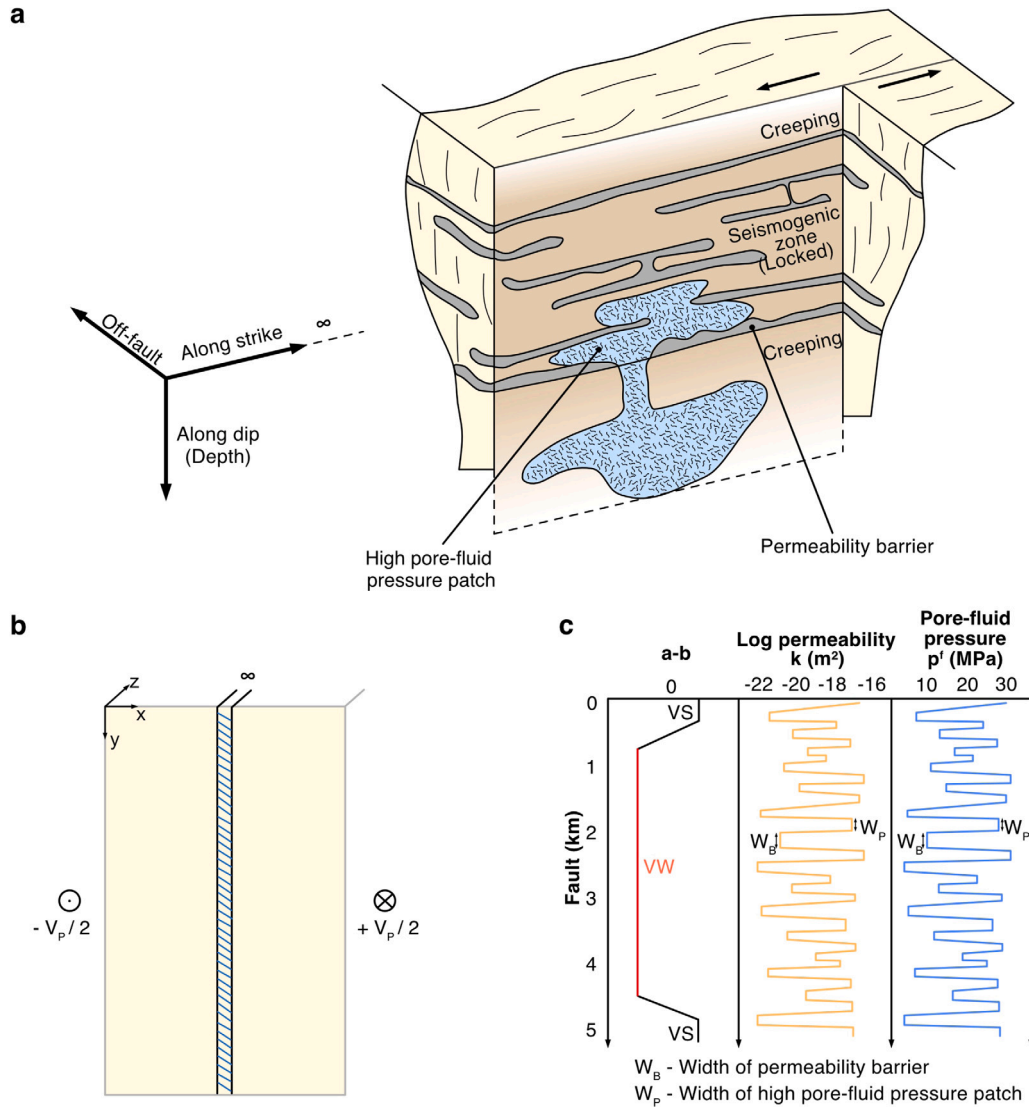


Fig. 1. Model setup. (a) solid–fluid interaction in the fault zone. Gray stripes indicate permeability barrier — structure with low permeability; Blue zone represents patches of fluids with high pore-fluid pressure (Modified after [Ross et al., 2020](#)). (b) 2D fault anti-plane model, loading with velocity $V_p/2 = 10^{-9}$ m/s. (c) Profile of frictional parameter (a–b), permeability and pore-fluid pressure on the fault. W_B and W_P are the width of permeability barriers and high pore-fluid pressure patch, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where a is rate-and-state frictional parameter, p_{eff} is effective pressure, $c_s = 3.1$ km/s is shear wave velocity, $\mu = 30$ GPa is shear modulus. Based on our models, velocity weakening parameter $a = 0.005$. Given that fluid pressure (effective pressure) varies dynamically during simulation, we use an average effective pressure of $p_{\text{eff}} = 20$ MPa, resulting in a threshold velocity $V_{\text{th}} = 2 \times 10^{-2}$ m/s. Therefore, a seismic event is defined when the maximum slip rate exceeds the threshold velocity $V_{\text{max}} > V_{\text{th}}$, whereas an aseismic event propagates with a maximum velocity lower than the threshold velocity, $V_{\text{max}} < V_{\text{th}}$.

3. Results

3.1. Role of permeability barriers and high pore-fluid pressure patches in fault slip dynamics

We conduct a parameter-space study by varying widths of permeability barriers (W_B) and high pore-fluid pressure patches (W_P), relative to the nucleation size $h^* = 2\mu b l / \pi(b - a)^2 p_{\text{eff}}$. Here, a , b and l are frictional parameters of rate- and state-friction law, μ is shear modulus and p_{eff} is effective pressure. The nucleation size describes the

critical size or length scale of a slipping region (or crack) at which it becomes unstable and can transition into dynamic rupture or runaway propagation ([Rice and Ruina, 1983](#); [Rubin and Ampuero, 2005](#)). Across all models, while keeping $h^* = 91.36$ m, we vary W_P in the range of 22.84–182.72 m and W_B within 45.68–91.36 m. These ranges maintain W_B/h^* between 0.5 and 1 and W_P/h^* between 0.4 and 2. Sampling these ratios allows us to explore how fault behavior changes as the pressurized and barrier zones transition from localized features to wider structures. These configurations are simplified and not meant to represent a uniquely realistic scenario of permeability and pore-fluid pressure in natural fault zones. Instead, they serve as reference cases to isolate the effects of permeability barriers and localized high pore-fluid pressure patches on slip behavior. This approach allows us to systematically investigate how the width of these heterogeneities influences fault slip. Therefore, this setup provides a baseline for comparison with models that include permeability evolution in the following sections, in which the feedback between permeability, pore-fluid pressure and fault slip is fully accounted for. A summary of the modeling parameters is available in [Table 2](#).

In this first set of models, to isolate the effects of W_B and W_P on seismic and aseismic slip behaviors, permeability barriers and high

Table 1
Model parameters.

Parameter	Symbol	Value
Numerical domain		
Length in x direction	L_x	2 km
Length in y direction	L_y	5 km
Resolution in x direction	d_x	0.02 m
Resolution in y direction	d_y	7.68 m
Material		
Shear modulus	μ	30 GPa
Solid bulk modulus	K^s	50 GPa
Effective pressure	p^f	min/max 5–30 MPa
Fluid pressure	p^f	min/max 10–25 MPa
Solid density	ρ^s	3000 kg/m ³
Fluid density	ρ^f	1000 kg/m ³
Solid compressibility	β^s	2.5×10^{-11} 1/Pa
Fluid compressibility	β^f	4×10^{-10} 1/Pa
Solid viscosity	η^s	10^{25} Pa s
Fluid viscosity	η^f	10^{-3} Pa s
Gravity	g	0 m/s ²
Critical nucleation size	h^*	91.36 m
Cohesive zone size	Λ_0	27.9 m
Porosity	ϕ	1%
Loading velocity	V_p	2×10^{-9} m/s
Permeability evolution		
Initial reference permeability	k^*	10^{-18} – 10^{-21} m ²
Healing time	t_k	10 h–10 years
Characteristic slip distance	l_k	0.1–1 m
Stress sensitivity parameter	σ^*	30 MPa
Maximum permeability	$k_{\max}$	10^{-15} m ²
Minimum permeability	$k_{\min}$	10^{-21} m ²
Rate- and state-friction		
Direct parameter	a	0.005
Evolutionary parameter	b	0.019
Reference velocity	V_0	10^{-9} m/s
Characteristic slip distance	D_c	0.6 mm
Reference friction	f_0	0.6

pore-fluid pressure patches are assumed to be time-invariant. In this setup, all barrier patches share a uniform width W_B , and all high pore-fluid-pressure patches share a uniform width W_P . We identify two key non-dimensional parameters: the ratio of permeability barrier width to nucleation size W_B/h^* , and the ratio of high pore-fluid pressure patch width to nucleation size W_P/h^* . These parameters modulate the hydro-mechanical structure of the fault and, consequently, its shear strength $\tau_s = f \cdot p_{\text{eff}} = f \cdot (p^t - p^f)$, which is controlled by both the friction coefficient f and the effective pressure p_{eff} (the difference between total pressure p^t and pore-fluid pressure p^f). Our results show that while W_P/h^* remains between 0.5 and 1, increasing W_B/h^* from 0.4 to 2 transitions the fault behavior from slow slip and aseismic creep to cascades of seismic and aseismic slip, ultimately leading to full seismic rupture (Fig. 2a). This indicates that wider permeability barriers promote fault instability and favor conditions under which dynamic rupture may emerge.

In the model M1 with the largest W_B (orange line in Fig. 2b), the slip rate surpasses the threshold velocity $V_{th} = 0.02$ m/s, as predicted analytically in Eq. (4). The only exception is the first patch, where slip rate goes below the threshold, because the rupture is still in the raising/growing phase. This behavior is illustrated in Fig. 2c, where a seismic event nucleates on a locked permeability barrier patch and then propagates through creeping patches of high pore-fluid pressure. These creeping patches play a dual role in fault dynamics. Firstly, they concentrate stress at the boundaries of locked zones (permeability barriers in Fig. 2c zoom-in view), promoting localized rupture initiation. Secondly, they act to modulate the propagation of slip fronts, which may decelerate, cascade, or arrest depending on the width of the surrounding high-pressure zones. Depending on the width of these creeping patches, the rupture can either propagate continuously, exhibit start-and-stop behavior (cascades), or arrest.

Table 2
Parameter list of Model M1–M6.

Model	W_P (m)	W_B (m)	h^* (m)	l_k (m)	t_k
M1	180	80	91.36	–	–
M2	83	65	91.36	–	–
M3	43	65	91.36	–	–
M4	40–110	50–130	91.36	–	–
M5	40–110	50–130	133.84	1	10 h
M6	40–110	50–130	133.84	1	1 year

Model M2, with medium W_B , exhibits transitional behavior, combining both seismic and aseismic slip modes. Here, the maximum slip rate exceeds the threshold velocity during each seismic event, only to fall below it in the post-seismic phase (green line in Fig. 2b). As the size of permeability barriers in Model M2 is smaller than in Model M1, each seismic event decelerates sooner due to the stability of patches of high pore-fluid pressure. These stable patches transfer less stress to neighboring barriers, triggering a continuous cascade of smaller seismic events along the fault (Fig. 2d). When W_B is at its minimum value, Model M3 reveals only slow slip coupled with aseismic creeping, as the slip rate never crosses the seismic threshold (blue line in Fig. 2b). W_B in this scenario is insufficient relative to the nucleation size to trigger seismic instability.

In summary, low-permeability barriers inhibit fluid infiltration, so fluids tend to bypass these regions rather than penetrate them. As a result, pore-fluid pressure inside barriers remains low, effective normal stress stays high, and these segments behave as seismic asperities that concentrate stress and host dynamic rupture. In contrast, high-permeability patches promote fluid percolation, leading to elevated pore-fluid pressure, reduced effective normal stress, and predominantly aseismic slip. This aseismic slip, however, redistributes stress and can load adjacent low-permeability barriers, ultimately triggering earthquakes there. Thus, the fault expresses a continuum of seismic and aseismic behaviors controlled by the along-dip widths of barriers (W_B) and high pore-fluid-pressure patches (W_P) relative to the nucleation length (h^*).

3.2. Permeability evolution and its role in fault slip dynamics

While our initial models use a simplified, static arrangement of permeability barriers to isolate their effects, we also explore a more realistic, heterogeneous configuration. In this extended setup, low- and high-permeability zones are assigned along strike using a stochastic procedure, so that W_B and W_P follow a spatially irregular distribution that mimics the strong small-scale heterogeneity of natural fault zones, while remaining fixed in time (Table 2, Model M4). This added heterogeneity captures the hydraulic variability observed in natural systems and reproduces a broader range of slip behaviors, some of which resemble clustered sequences observed in nature, with varying sizes and aseismic slip in multiple directions (Fig. C.1a). By integrating features of the simplified models (M1, M2, and M3), this approach demonstrates that permeability barriers and pore-fluid pressure patches fundamentally govern fault slip dynamics, giving rise to a spectrum of emergent behaviors that includes both fast and slow events, without implying distinct classes of rupture.

To better investigate the dynamic behavior of permeability and pore-fluid pressure in a heterogeneous fault, we incorporate a recently adopted permeability evolution law (Eq. (3b)) (Zhu et al., 2020), which captures fault weakening and valving behavior consistent with geological observations (Sibson, 1990, 1992; Bense et al., 2013). In our study, to assess the sensitivity of fault slip behavior to permeability change, we test a broad range of healing time t_k values, ranging from 10 h to 10 years, while keeping other parameters constant. In these simulations (Model M5 and Model M6), W_B and W_P are assigned random widths within $W_B = 40$ –110 m and $W_P = 50$ –130 m, while keeping $h^* =$

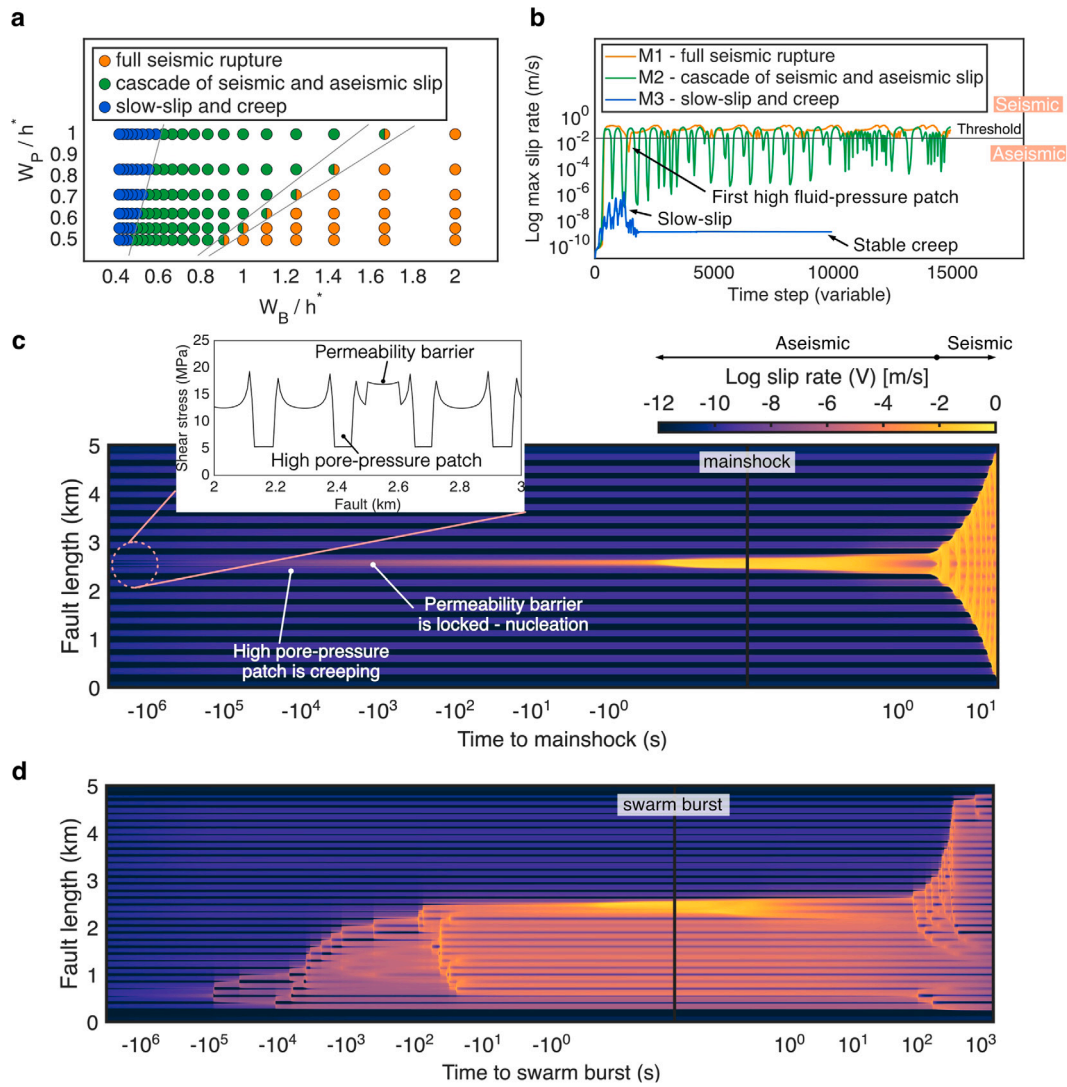


Fig. 2. Impact of width of barriers and patches on slip pattern. (a) Comparison between W_B/h^* and W_P/h^* . (b) Maximum slip rate in log scale for models with different slip patterns. (c) Slip rate in log scale vs. time to mainshock for Model M1, showing seismic event nucleates on locked permeability barrier and accelerated by the surrounding creeping patches, leading up to full seismic rupture on the fault. The mainshock is defined as the event with largest magnitude. The zoom-in view shows stress concentration at the boundary of permeability barriers. (d) Slip rate in log scale vs. time to onset of swarm-like burst for Model M2, producing cascade of seismic and aseismic slip. Swarm-like burst refers to a rapid sequence of small to moderate earthquakes, occurring in a localized area over a short period of time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

133.84 m. The initial conditions for Model M5 and M6 are identical except for the t_k .

Our simulations show that the healing time t_k controls how fluids redistribute and, in turn, how slip localizes on the fault. After the first rupture events have damaged and unsealed the fault, longer healing times keep these damaged zones permeable for longer. In this high-permeability state, fluid can migrate efficiently away from initially over-pressurized patches, so rapid pressure equilibration keeps pore-fluid pressure relatively higher and with modest contrasts. Although this pressure state is more homogeneous, it sustains reduced effective pressure over broad fault regions, thus promoting prolonged aseismic slip on and around previously ruptured areas. This behavior limits the growth of individual dynamic events. In contrast, when t_k is short, permeability recovers quickly and fluid pathways reseal soon after each event, trapping fluids between permeability barriers. This inefficient drainage leads to strong local pore-pressure build-up, sharp effective-stress gradients, and stress concentration that favors runaway, system-size dynamic ruptures spanning the entire unstable fault segment. For example, Model M5 (Fig. 3a,c,e) and Model M6 (Fig. 3b,d,f)

demonstrate these distinct behaviors in simulations of seismic cycles. Model M5 (Fig. 3a), with $t_k = 10$ h, simulates rapid recovery of increased permeability over several seismic cycles, resulting in minimal fluid mobility within the fault throughout the cycles. At the beginning of the cycle, seismic clusters emerge at kilometer 4 along the fault, then propagate upwards and culminate in the largest event of the cycle. For descriptive purposes, we refer to this largest-moment event as a “mainshock”. This large event propagates through the entire fault until complete rupture. Preceding this mainshock, foreshocks occur at multiple time scales, ranging from days to seconds. The first foreshock arises approximately 1 day before the mainshock, followed by quiet intervals of hours and then clusters of seismic events minutes before the largest rupture, suggesting a progressive cascade of failure. Each event nucleates at the boundary of the previous ones and remains creeping during the inter-seismic period. The creeping accelerates after each seismic event compared to the previous one during the foreshock sequences, leading up to the mainshock (Fig. 3c). This acceleration of slip indicates a scaling between maximum slip rate $V_{\max}$ and time t as $V_{\max} \propto t$, similar to the studies of precursory foreshocks on rough faults (Cattania and Segall, 2021).

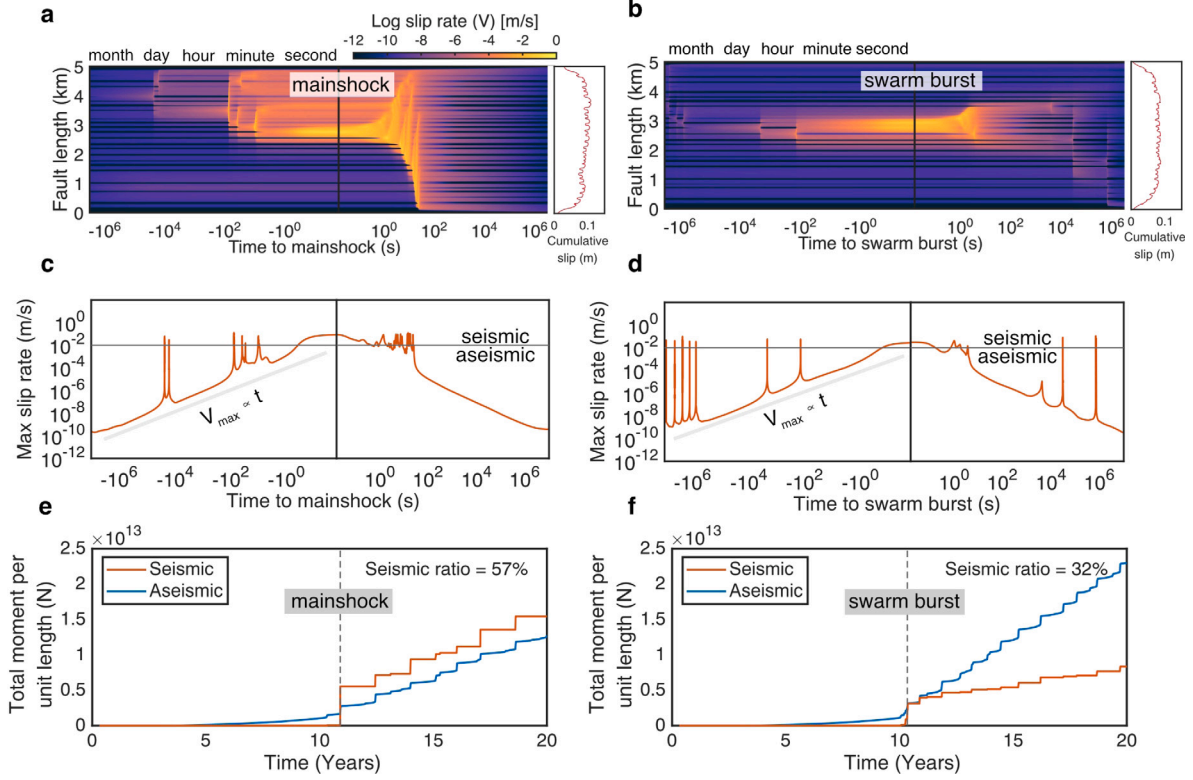


Fig. 3. Slip rate evolution and moment release. (a), (c) and (e) Model M5, in which the healing time t_k is 10 h, shows system-size dynamic rupture. (b), (d) and (f) Model M6, in which the healing time t_k is 1 year, represents swarm-like events. (a) Slip rate evolution on the fault in time for Model M5, showing the slip rate acceleration during foreshocks sequences, leading to the mainshock and complete dynamic rupture on the fault. Right panel shows the accumulative slip. (b) Slip rate evolution on the fault in time for Model M6, showing active swarm-like burst. (c) Maximum slip rate vs. time to the end of the mainshock in (a), showing slip acceleration until the mainshock. The horizontal line represents the threshold slip rate for seismic or aseismic slip. The gray slope shows the scaling between maximum slip rate $V_{\max}$ and time t as $V_{\max} \propto t$. (d) Maximum slip rate vs. time to the end of the swarm burst in (b). (e) and (f) Moment release of both seismic and aseismic over the sequences over 20 years, with seismic ratio (seismic moment/total moment) = 57% for M5, and 32% for M6.

A longer healing time changes the behavior of the fault system due to the slower recovery rate of permeability. Model M6 (Fig. 3b), with $t_k = 1$ year, simulates a scenario where the onset of seismicity allows unsealing of the permeability barriers, enhancing fault-zone fluid percolation. Due to the slow healing time, pore-fluid pressure redistributes efficiently, causing the propagation of seismicity as a spatially distributed sequence of small events, resembling observed swarms. Unlike Model M5, seismic clusters nucleate at the bottom of the fault but do not trigger a system-size rupture; instead, they propagate along the fault while maintaining their characteristics as transient, clustered bursts of seismicity — that is, a rapid sequence of small to moderate earthquakes, occurring in a localized area over a period of time (~ 20 days). Despite the different slip patterns from Model M5, the resulting distributed slip bursts exhibit a similar scaling of $V_{\max} \propto t$, indicating that this scaling can emerge across diverse modes of slip behavior — from transient clustering to system-scale rupture — within a unified hydro-mechanical framework. This also indicates that slip acceleration can precede both large-scale rupture and distributed similar-sized events, suggesting that these outcomes exist along a continuum, shaped by fault hydromechanics and cannot be distinguished based on this scaling behavior alone. We discuss further conditions necessary to determine whether a fault slip results in similar-sized events or evolves into a system-size rupture in the following section.

To understand the partitioning of seismic and aseismic slip, both spatially and temporally, we calculated the seismic and aseismic moment release throughout several seismic cycles. This provides an energy-based estimation released by the fault. The moment release

can be calculated as $M_0 = \mu AD$, where D represents average slip and A is the rupture length (in our case, it is 1D). Seismic moment is computed when slip rate exceeds the seismic/aseismic threshold (as defined in Section 2), and aseismic moment is calculated when it falls below. Although the cumulative slip (or total moment) is comparable in both models (Fig. 3a,b), our results show distinct behaviors in the seismic cycles spanning 20 years. In Model M5, characterized by a smaller t_k , earthquake sequences dominate in the seismic cycle (Fig. C.1b). Each large event within the cycle produces more slip (Fig. 3a right panel) and generates significantly more seismic moment than the aseismic moment released (Fig. 3e), due to the extended rupture length and slip. In particular, after the mainshock, the cumulative seismic moment is above the cumulative aseismic moment, with a seismic ratio (seismic moment/total moment) of 57%. In contrast, Model M6, characterized by a larger t_k , produces spatially distributed sequences of slip events that lack a dominant rupture and instead involve transient, similar-sized bursts of activity along the fault (Fig. C.1c). The seismic moment associated with these distributed slip bursts is lower than the aseismic moment released elsewhere on the fault. As a result, the cumulative seismic moment consistently falls below the cumulative aseismic moment, with a seismic ratio of 32% at the end of the cycles (Fig. 3f). These results thus highlight the critical role of permeability evolution and healing time in governing the emergence of diverse fault slip behaviors. Depending on hydro-mechanical conditions, the system may produce sequences that qualitatively resemble swarms or mainshock-aftershock patterns, without implying that these represent fundamentally distinct rupture modes.

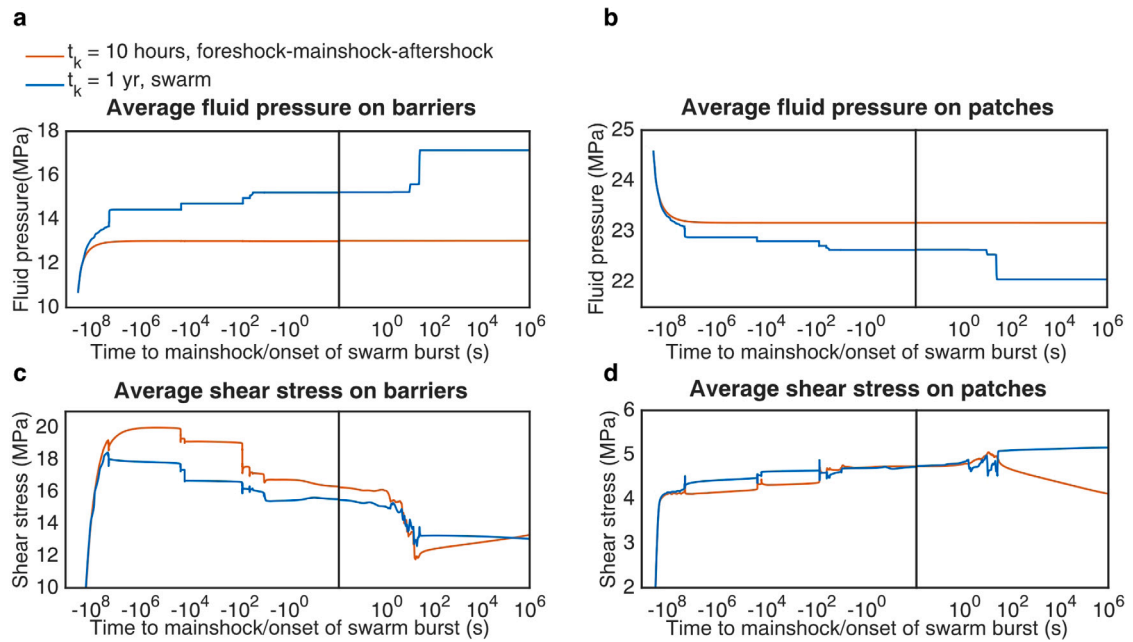


Fig. 4. Evolution of fluid pressure and shear stress. Average fluid pressure on (a) permeability barriers and (b) patches of high pore-fluid pressure vs. time to mainshock or onset of swarm burst, showing the comparison between model with $t_k = 10$ h (red line) and with $t_k = 1$ yr (blue line). Average shear stress on (c) permeability barriers and (d) patches of high pore-fluid pressure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3. Conditions for small similar-sized events or large system-size rupture

Our results identify the key parameters that govern the spectrum of fault slip behaviors, ranging from distributed small similar-sized sequences to large system-size rupture. Firstly, similar-sized events emerge when the characteristic widths of permeability barriers and high pore-fluid-pressure patches are of comparable order, that is, when the average W_B/h^* and W_P/h^* are similar. In Models M5 and M6, we set $W_B/h^* = 0.29\text{--}0.82$ and $W_P/h^* = 0.37\text{--}0.97$. When W_B/h^* and W_P/h^* are of similar magnitude, rupture tends to remain distributed and does not immediately grow to system scale.

Another crucial factor is how efficiently fluids can redistribute along the fault. The efficiency of fluid redistribution is primarily controlled by the healing time t_k . For a small t_k (10 h), permeability reseals quickly after each event, reverting from its damaged value back to the low background level between cycles. For a large t_k (1 year), this recovery is much slower. Consequently, damaged, high-permeability zones persist over multiple events. The change in permeability from minimum to maximum becomes more permanent. Fluids can drain from over-pressurized patches into neighboring barriers. On average, this leads to increasing pore-fluid pressure in barriers (blue line in Fig. 4a) and decreasing pressure in high pore-pressure patches (blue line in Fig. 4b), consistent with diffusion from high- to low-pressure regions. This phenomenon, known as fluid diffusion, contrasts with the scenario of a small t_k , where increased permeability is followed by an immediate decrease. In this case, fluid remains relatively static within the fault, as indicated by constant fluid pressure in both permeability barriers and high pore pressure patches (red line in Fig. 4a–b).

Generally, slow or minimal fluid diffusion preserves high pressure contrasts across fault zones, maintaining conditions that favor localized slip. In Fig. 5a, fluid pressure in barriers remains relatively unchanged from the start of the cycle until the mainshock. As clustered events propagate along the fault, a larger rupture nucleates at 2.75 km once sufficient stress accumulation has occurred across multiple barriers. Conversely, faster fluid diffusion smooths out the pressure contrast across permeability barriers. In Fig. 5b, before the seismic burst, the permeability barrier at 3.3 km is unsealed, allowing fluids to penetrate

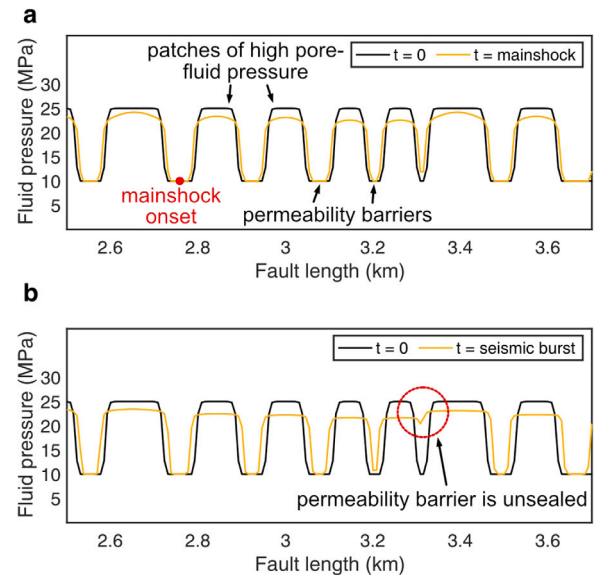


Fig. 5. Snapshot of fluid pressure at $t = 0$ and $t = \text{mainshock}$ or seismic burst in the fault position 2.5–3.7 km. (a) Fluid pressure comparison between $t = 0$ and $t = \text{mainshock}$ for Model M5, where rupture propagates forwards along the fault until the onset of system-size rupture (mainshock) at around 2.8 km. (b) Fluid pressure comparison between $t = 0$ and $t = \text{seismic burst}$ for Model M6, showing the unsealing of permeability barriers before the onset of seismic burst, which stabilizes the fault.

more easily from the high pore-pressure patch to this barrier due to its shorter length. This smoothing effect results in the widening of the high pore-pressure patch and shortening of barriers, illustrating a shift from dominantly seismic to dominantly aseismic response within the parameter space (Fig. 2a). Consequently, more aseismic slips build up, leading to a reduction in average shear stress (Fig. 4c) on permeability barriers by 1–3 MPa, compared to the static case. While the shear stress

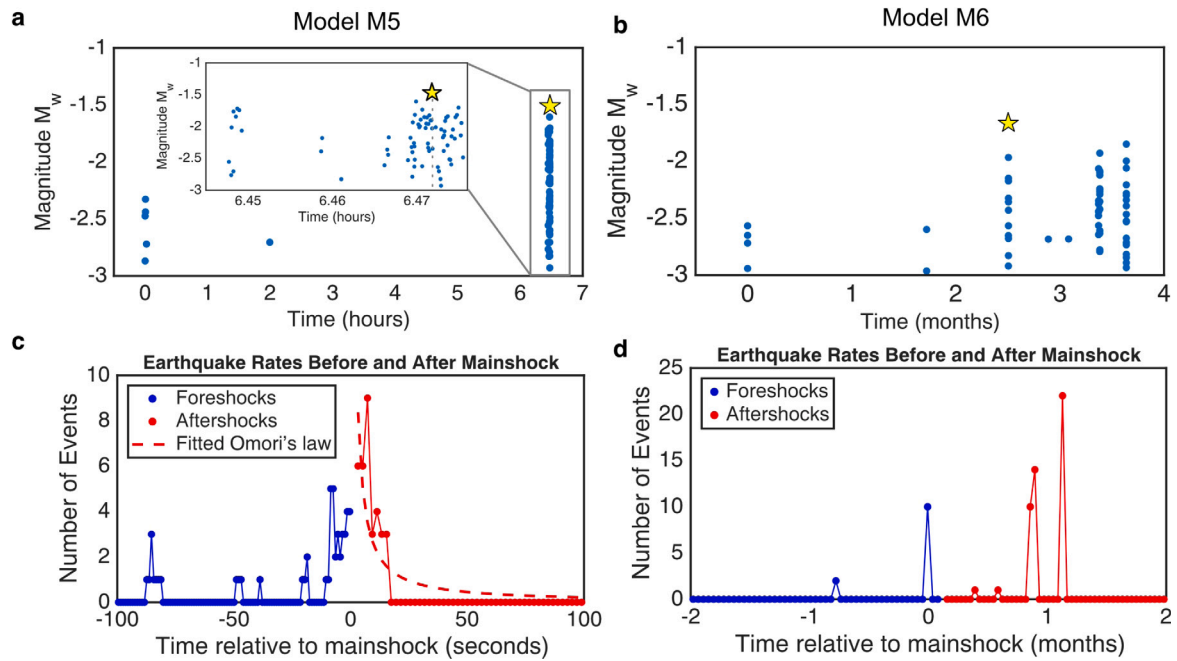


Fig. 6. Catalog and earthquake rates. Catalog calculated for each event in M_w for (a) Model M5 and (b) Model M6. The yellow star represents the system-size rupture (mainshock), which is defined as the event with largest magnitude. (c) and (d) shows the average rate of foreshocks and aftershocks is shown as a function of time for Model M5 and Model M6, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

on high pore-pressure patches remains comparable over time (Fig. 4d), the decreased shear stress on permeability barriers stabilizes the fault locally, impeding large-scale rupture propagation and reducing overall seismic moment release.

4. Discussions

4.1. Comparison of models with observations and its implication

Our models simulate a range of slip behaviors that resemble those observed in nature, emerging from interactions between hydraulic heterogeneity and permeability evolution. As shown in Model M5, when the fault is characterized by a heterogeneous hydraulic structure, with low-permeability barriers embedded within regions of elevated pore-fluid pressure, fluid diffusion along the fault is strongly limited. This leads to migratory slip behavior, where direct triggering between permeability barriers (asperities) is mediated by accelerated creep within high-pressurized fluid patches. This interaction facilitates the successive failure of neighboring patches, triggering cascades of accelerating slip that can culminate in a larger dynamic rupture. The resulting sequence shares characteristics with FMA behavior observed in natural faults, whose temporal decay rates align with Omori's law (Fig. 6a,c). Such phenomena have been observed in the 2004 Mw 6 Parkfield earthquake (Khoshmanesh and Shirzaei, 2018) and 2009 Mw 6.3 L'Aquila earthquake (Sugan et al., 2014). Conversely, when permeability barriers are sparse relative to high pore-fluid pressure patches, fault is stabilized and aseismic creep is dominant (Model M3). When these permeability barriers are unsealed, the increased permeability facilitates fluid redistribution along the fault through diffusion (Model M6). Rapid fluid diffusion smooths out pore-fluid pressure heterogeneity in space, driving rapid sequences of spatially clustered events, whose temporal decay departs from Omori's law and shares features with observed earthquake swarms (Fig. 6b,d), as observed in California (Ross et al., 2020; Shelly et al., 2016; Li et al., 2021; Shelly et al., 2015), in the Corinth Gulf of Greece (De Barros et al., 2020) and 2020–2024 Noto sequences (Yoshida et al., 2023).

In nature, swarms can act as precursory foreshocks and potentially anticipate a mainshock under favorable fluid conditions — specifically when the fault is pressurized by high pore-fluid patches. Such a pressurization arises when fluids can diffuse efficiently along certain pathways but are impeded in other directions by spatial variations in fault-zone permeability. This entrapment creates localized over-pressurized zones, which can promote seismicity and, under favorable stress conditions, contribute to larger earthquakes. These dynamic fluid conditions could explain the occurrence of many observed earthquake sequences (Shelly, 2024; Bachura et al., 2021; De Barros et al., 2020). For example, the 2016–2017 Central Italy seismic sequence provides a clear example, as high-resolution seismic data (Michele et al., 2020) reveal a high degree of complex fault structures where fluids play a key role in controlling seismicity. This sequence indicates the (re)activation of different fault segments throughout a series of M_w 5.9–6.5 events, characterized by migratory behavior. This observation supports the proposed mechanism, where dynamic fluid interactions control the local stress of specific fault patches. This leads to both distributed and larger-scale ruptures, reflecting the continuum of slip behaviors that do not always conform to typical FMA sequences.

In the models, increasing the healing time t_k from 10 h to 10 years in our simulations decreases the seismic ratio from 57% to 32% (Fig. C.2). This may represent one plausible mechanism contributing to the variability in seismic ratio, which ranges from 3% to 70% in shallow slow slip events (<10 km) across various tectonic settings (Passarelli et al., 2021). Additionally, the migration velocity of seismicity decreases with increasing t_k . For example, Model M5, with $t_k = 10$ h, exhibits a foreshock migration rate of 0.1 km/day (Fig. 7a), closely matching the rate ($V_m = 0.5$ km/day) observed in the foreshock sequence in 2009 Mw 6.3 L'Aquila earthquake (Sugan et al., 2014). Such a rapid migration is often interpreted as evidence for aseismic creep (Lohman and McGuire, 2007; Chen and Shearer, 2013). Conversely, Model M6, with $t_k = 1$ year, shows a slower migration rate for 1.5 months (4×10^6 s) before acceleration (Fig. 7b). We note that the sequence of distributed slip bursts in Model M6 does not closely follow the classical simple diffusion curve, consistent with the non-linear nature of fluid diffusion and the spatial and temporal variability in permeability (Ross et al., 2020). A

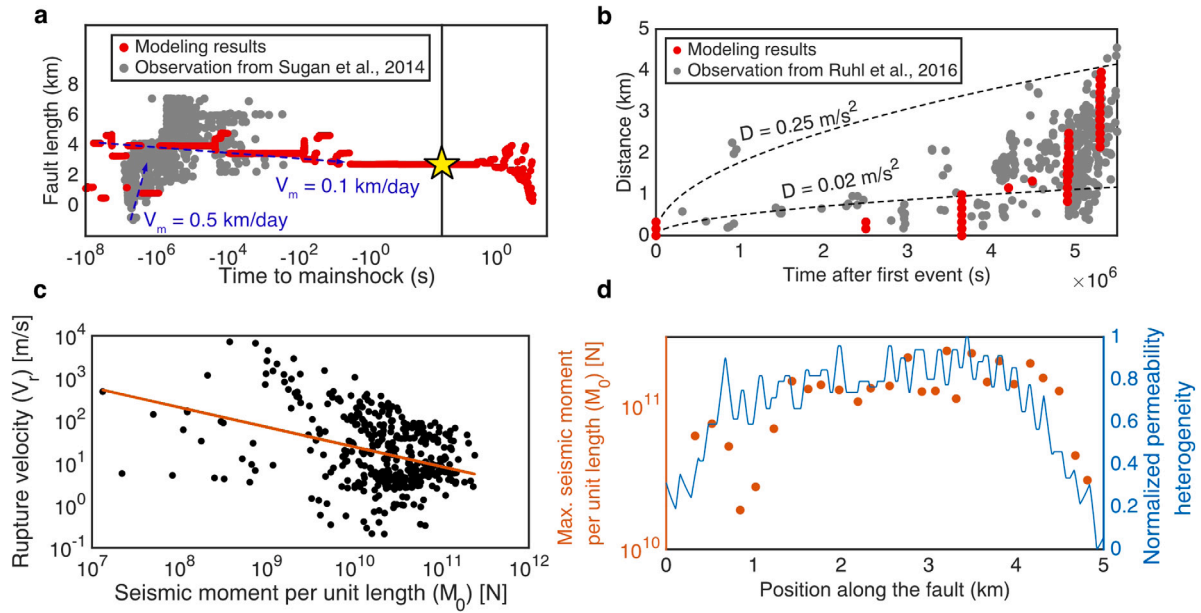


Fig. 7. Comparison of migration and scaling properties between the numerical experiments and observations. (a) Spatial-temporal evolution of earthquake sequence in Model M5, compared to the foreshock sequence in 2009 Mw 6.3 L'Aquila earthquake (Sukan et al., 2014). Each dot represents an event, with the yellow star indicating the mainshock in both the model and the observation. The blue dash lines show the migration of foreshock with velocity of V_m . (b) Spatial-temporal evolution of swarm-like burst in Model M6, compared to the swarm migration in 2008 Mogul (Nevada) earthquake swarm (Ruhl et al., 2016). The radial distance of seismicity relative to the inferred injection point along with several diffusivity curves following $r = \sqrt{4\pi Dt}$ (Shapiro et al., 1997). These curves provide the range of diffusivity of $D = 0.02$ – 0.25 m/s², matching with the observed swarms associated with fluid diffusion. (c) Moment-rupture velocity scaling implies inverse proportional relationship. Rupture velocity, also known as rupture speed, is taken at the rupture front of each event. (d) Maximum seismic moment M_0 in each permeability barrier on the fault, compared to the normalized permeability heterogeneity. The normalized permeability heterogeneity represents the intensity of heterogeneity in permeability, where 0 indicates a smooth permeability distribution and 1 indicates a distribution with high perturbation. The value is derived from the normalized coefficient of variation in permeability. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

close match with the observation of the 2008 Mogul (Nevada) seismic swarm (Fig. 7b) (Ruhl et al., 2016) also highlight that spatio-temporal migration of earthquakes show additional finer scale intricacies than those predicted by a simple diffusivity model. Nonetheless, on the longer duration of the swarm, the predicted seismic diffusivity of $D = 0.02$ – 0.25 m/s² of our model is within the range associated with fluid flow observed in the Mogul swarm (Ruhl et al., 2016). This suggests that even if long-time scales are compatible with diffusion, shorter time scale variabilities in swarm-like migration patterns are dominated by time-varying permeability predicted by our model. These contrasting mechanisms highlight how fluid migration conditions — modulated by evolving permeability — govern the style, spatial distribution, and productivity of fault slip. The choice of k_{max} may also influence the seismicity migration (Marguin and Simpson, 2023), but this effect is not systematically explored here.

While our models explore various magnitudes of t_k , constraining t_k in natural fault zones is challenging due to the lack of direct measurements of permeability recovery. The recovery rate can also vary at different locations on the fault. Besides t_k , the permeability enhancement evolution distance l_k is another critical parameter influencing permeability evolution. In our models, varying l_k within the range of 0.1–1 m does not significantly affect the seismic ratio (Fig. C.2). Despite experimental and theoretical advancements on permeability changes (Zhu et al., 2020; Ozawa et al., 2024), compaction and dilation (Marone et al., 1990; Segall and Rice, 1995; Proctor et al., 2020; Dal Zilio et al., 2022a), these results call for further analyses to better understand the impact of fluid flow on the entire fault volume, both on and off the fault.

4.2. Scaling relationship

Although our model adopts a simplified setup, the simulated slip events reproduce scaling trends observed in small earthquakes and

tremor- or swarm-like seismicity. As shown in Fig. 7c, the scaling indicates that rupture velocity (speed) V_r is inversely proportional to seismic moment, consistent with observations in shallow fault systems (Passarelli et al., 2021). Slower rupture velocity in small earthquakes in shallow fault systems might result from increased fault heterogeneity, such as the distribution of permeability and pore-fluid pressure (Gao et al., 2012; Passarelli et al., 2021; Danré et al., 2024), which limits faster slip propagation, as demonstrated in fluid injection experiment in laboratory on rough permeable fault (Wang et al., 2024). Conversely, fast rupture may nucleate in areas with lower heterogeneity and a more homogeneous stress distribution. Further analysis of permeability structure on the fault also indicates that the heterogeneity of permeability is closely linked to seismic moment (Fig. 7d). In asperity models, faults are conceptualized as networks of small and strong patches, known as asperities, embedded within weaker and more compliant regions. These heterogeneities in fault strength and permeability influence seismic behavior, with the failure of individual asperities generating small earthquakes and controlling the overall seismic moment release (Ito et al., 2007; Passarelli et al., 2021). Other scaling relationships, including those between seismic moment and rupture length ($L \propto M_0^{1/3}$) and duration ($T_d \propto M_0^{1/2}$) (Fig. C.3), are consistent with observations in shallow thrust earthquakes (Ide et al., 2007; Denolle and Shearer, 2016), suggesting that permeability-controlled slip spans the continuum between slow and regular earthquakes.

4.3. Model limitations

In the presented models, we have revolved the processes of seismic and aseismic slip in detail. However, the high spatial and temporal resolution required for this level of analysis comes at a substantial computational cost. As a result, our 2D modeling framework naturally limits the number of simulated seismic events. The total event count

is therefore lower than what is typically observed in nature. For the same reason, the seismic moment per unit length produced in our simulations, ranging from $10^8 - 10^{11}$ N, is also smaller than values inferred from natural earthquakes. To fully assess the robustness of the modeled slip behaviors and scaling relationships, future work will require 2D or 3D simulations that can accommodate larger rupture domains and more realistic geometries.

Another limitation is the simplification of the fault zone geometry. Although we incorporate heterogeneity through randomly distributed permeability and pore-fluid pressure, natural fault zones are considerably more complex. They commonly include a localized fault core surrounded by a fractal-like damage zone (Ben-Zion and Sammis, 2003), features that cannot be fully captured within our current setup. Despite these simplifications, the model is able to isolate the fundamental roles of permeability barriers and pore-fluid pressure patches, as well as the dynamic interactions between them. These factors control on the initiation and evolution of seismic and aseismic slip, providing insights that can guide for future challenging modeling efforts.

5. Conclusions

This study highlights the key role of permeability barriers and pore-fluid-pressure variations in shaping the onset and evolution of both seismic and aseismic slip. Using a fully coupled poro-visco-elasto-plastic fault model with rate-and-state friction and evolving permeability, we show how fault-zone hydromechanics can generate a broad spectrum of slip behaviors. A summary of results is illustrated in Fig. 8. By varying the along-strike widths of low-permeability barriers (W_B) and high pore-fluid-pressure patches (W_P) relative to the nucleation size (h^*), we demonstrate that hydraulic heterogeneity alone can produce behaviors ranging from predominantly aseismic creep to cascades of smaller events and system-size dynamic rupture. Low-permeability barriers inhibit fluid infiltration, maintain low pore-fluid pressure and high effective normal stress, and thus behave as seismic asperities that concentrate stress and host dynamic rupture. In contrast, high-permeability, over-pressurized patches favor aseismic slip, but this slip redistributes stress and can load neighboring barriers, ultimately triggering earthquakes there.

When permeability is allowed to evolve with slip and healing, the characteristic healing time t_k emerges as a second key control on sequence style. Short healing times reseal damaged pathways quickly, limit fluid redistribution, trap fluids between barriers, and promote strong stress concentrations that favor system-size ruptures and higher seismic ratios. Long healing times keep damaged zones permeable over multiple events, enhance along-fault fluid diffusion, reduce pore-pressure contrasts, and favor swarm-like, distributed sequences in which aseismic moment dominates. By comparing these regimes with observed seismicity patterns, our results provide a hydro-mechanical framework in which FMA-like sequences and earthquake swarms arise from the same underlying physics, rather than from fundamentally distinct triggering mechanisms. Ultimately, our findings support the view that seismic and aseismic slip patterns are not discrete categories, but emerge from continuous variations in fault hydro-mechanical properties, modulated by permeability structure, permeability evolution, and pore-fluid-pressure redistribution.

CRedit authorship contribution statement

Jiayi Ye: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Luca Dal Zilio:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Funding acquisition, Conceptualization. **Luigi Passarelli:** Writing – review & editing, Writing – original draft, Validation, Investigation. **Domenico Giardini:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Governing equations

The physical representation of a deforming continuum in a static gravity field is characterized by the following set of conservation equations for total momentum (solid matrix and fluid, Eq. (A.1a)), fluid momentum (Eq. (A.1b)), compressible solid mass (Eq. (A.1c)), and compressible fluid mass (Eq. (A.1d)):

$$\nabla \cdot \sigma + g\rho^t = \rho^t \frac{D^s v^s}{Dt}, \quad (\text{A.1a})$$

$$v^D = -\frac{k}{\eta^f} \left(\nabla p^f - \rho^f \left(g - \frac{D^f v^f}{Dt} \right) \right), \quad (\text{A.1b})$$

$$\nabla \cdot v^s = -\frac{1}{K^d} \left(\frac{D^s p^t}{Dt} - \alpha \frac{D^f p^f}{Dt} \right) - \frac{p^t - p^f}{\eta^\phi (1 - \phi)}, \quad (\text{A.1c})$$

$$\nabla \cdot v^D = \frac{\alpha}{K^d} \left(\frac{D^s p^t}{Dt} - \frac{1}{B} \frac{D^f p^f}{Dt} \right) + \frac{p^t - p^f}{\eta^\phi (1 - \phi)}, \quad (\text{A.1d})$$

Superscripts s , f , t , and D denote solid, fluid, total matrix, and Darcy, respectively. σ is the Cauchy stress tensor, ϕ is the porosity, ρ^s is the solid density, v^s is the solid velocity vector, t is time, v^D is the Darcy flux vector ($v^D = \phi(v^f - v^s)$), k is the permeability, η^f is the fluid viscosity, p^t and p^f are the total and fluid pressures, respectively, K^d is the drained bulk modulus, η^ϕ is the effective visco-plastic compaction viscosity, v^f is the fluid velocity vector, α is the Biot-Willis coefficient, B is the Skempton coefficient, and g is the gravity vector. The total momentum equation (Eq. (A.1a)) assumes negligible differences in solid and fluid accelerations.

The drained bulk modulus (K^d), Biot-Willis coefficient (α), and Skempton coefficient (B) are defined as:

$$K^d = (1 - \phi) \left(\frac{1}{K^\phi} + \frac{1}{K^s} \right)^{-1}, \quad (\text{A.2a})$$

$$\alpha = 1 - \frac{\beta^s}{\beta^d}, \quad (\text{A.2b})$$

$$B = \frac{\beta^d - \beta^s}{\beta^d - \beta^s + \phi(\beta^f - \beta^s)}, \quad (\text{A.2c})$$

$$\beta^d = \frac{1 + (\beta^s K^\phi)}{K^\phi - (K^\phi \phi)} \quad (\text{A.2d})$$

where K^ϕ is the effective bulk modulus of pores, K^s is the solid bulk modulus, β^d is the compressibility of the solid skeleton, β^s is the compressibility of the solid phase, and β^f is the compressibility of the fluid phase. The stress tensor is decomposed into deviatoric and

Summary of results

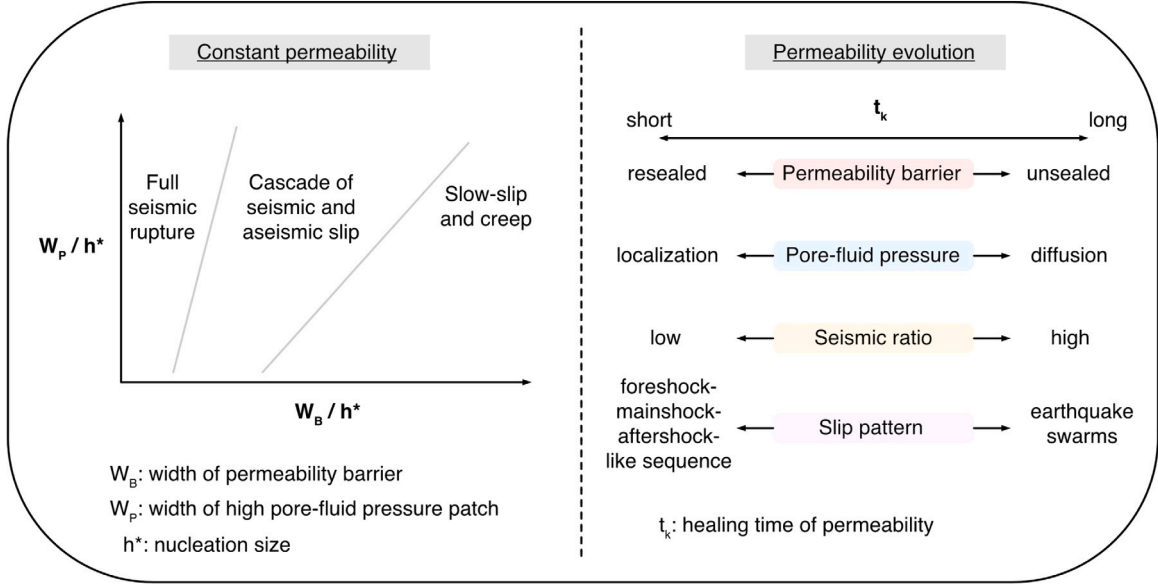


Fig. 8. Graphical summary of results.

volumetric components. The total pressure (p^t) and density (ρ^t) are coupled with porosity and computed from their respective fluid and solid matrix quantities:

$$p^t = p^f \phi + p^s(1 - \phi), \quad (\text{A.3a})$$

$$\rho^t = \rho^f \phi + \rho^s(1 - \phi), \quad (\text{A.3b})$$

where p^s and ρ^s are solid pressure and solid density, respectively. The effective bulk modulus of pores (K^ϕ) and effective visco-plastic compaction viscosity (η^ϕ) are computed as (Yarushina and Podladchikov, 2015):

$$K^\phi = \frac{2m}{1+m} \frac{\mu}{\phi}, \quad (\text{A.4a})$$

$$\eta^\phi = \frac{2m}{1+m} \frac{\eta_{s(vp)}}{\phi}, \quad (\text{A.4b})$$

where m is a geometrical factor, μ is the shear modulus, and $\eta_{s(vp)}$ is the effective visco-plastic shear viscosity of the solid matrix. For this study, m is set to 1 (cylindrical pores), reducing Eqs. (A.4a) and (A.4b) to μ/ϕ and $\eta_{s(vp)}/\phi$. The visco-plastic effective shear viscosity in Eq. (A.4b) is crucial in our rheological model.

The continuity equations for solid (Eq. (A.1c)) and fluid (Eq. (A.1d)) include poroelastic terms based on Biot's theory (Biot, 1941), allowing for the compressibility of the solid matrix and fluid, as well as viscous and elastic (de)compaction of the porous space. The full coupling of Eqs. (A.1c) and (A.1d) leads to compaction/decompaction processes, resulting in a fully coupled hydro-mechanical system.

Appendix B. Visco-elasto-plastic rheology

The deviatoric strain rate tensor components are defined under the assumption of infinitesimal strain as:

$$\dot{\epsilon}'_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) - \frac{1}{3} \text{Tr} \dot{\epsilon}_{ij}, \quad (\text{B.1})$$

where i and j are arbitrary indices and x_i and x_j are the 2-D spatial coordinates defining the tensor components. The total strain rate is a

combination of elastic, viscous, and plastic components:

$$\dot{\epsilon}'_{ij} = \dot{\epsilon}'_{ij}[\text{viscous}] + \dot{\epsilon}'_{ij}[\text{elastic}] + \dot{\epsilon}'_{ij}[\text{plastic}], \quad (\text{B.2})$$

The elastic component is defined as:

$$\dot{\epsilon}'_{ij}[\text{elastic}] = \frac{1}{2\mu} \frac{\bar{D}}{\bar{D}t} (\tau_{ij}), \quad (\text{B.3})$$

where the differential operator $\bar{D}/\bar{D}t$ represents the co-rotational time derivative of the Cauchy stress. Viscous deformation, following Newton's law, is defined using an effective shear viscosity of the bulk porous material (η_m):

$$\eta_m = \frac{\tau'_{II}}{2\dot{\epsilon}'_{II[v]}}, \quad (\text{B.4})$$

where τ'_{II} is the second invariant of the deviatoric stress, and $\dot{\epsilon}'_{II[v]}$ is the viscous component of the second invariant of the strain rate. Thus, the viscous contribution to the strain rate tensor is defined as:

$$\dot{\epsilon}'_{ij}[\text{viscous}] = \frac{\tau_{ij}}{2\eta_s}, \quad (\text{B.5})$$

where η_s is the shear viscosity. The fault is modeled as a finite-width shear zone, and plastic deformation is calculated as volumetric strain represented by a tensor, allowing plastic yielding to localize spontaneously.

The non-associated plastic flow law is defined through the plastic flow potential (Q), representing the mechanical energy per unit volume that supports plastic deformation (Vermeer, 1998):

$$Q = \tau_{II} - \sin(\psi)(p^t - p^f) - \cos(\psi)c. \quad (\text{B.6})$$

The plastic component of the deviatoric strain rate is:

$$\dot{\epsilon}'_{ij}[\text{plastic}] = \begin{cases} 0 & \tau_{II} < \tau_{\text{yield}} \\ \chi \frac{\partial Q}{\partial \tau_{ij}} = \chi \frac{\tau_{ij}}{\tau_{II}} & \tau_{II} = \tau_{\text{yield}} \end{cases}, \quad (\text{B.7})$$

with χ as the plastic multiplier ensuring that, if yielding occurs, the deviatoric stress always satisfies the yield criteria ($\tau_{II} = \tau_{\text{yield}}$). The

deviatoric components (τ_{ij}) of the stress tensor in Eq. (A.1a) are formulated using visco-elasto-plastic constitutive relationships and an implicit first-order finite-difference scheme in time:

$$\tau_{ij} = 2\eta_{vp} Z \dot{\epsilon}'_{ij} + \tau_{ij}^0 (1 - Z), \quad (\text{B.8a})$$

$$Z = \frac{\mu \Delta t}{\mu \Delta t + \eta_{vp}}, \quad (\text{B.8b})$$

where Z is the visco-elasticity factor, Δt is the computational time step, and η_{vp} is the effective visco-plastic viscosity characterizing plastic deformation intensity:

$$\eta_{vp} = \eta_m \frac{\tau_{II}}{2\eta_m \dot{\epsilon}_{II[p]} + \tau_{II}}, \quad (\text{B.9})$$

where η_m is the effective shear viscosity of the bulk porous material:

$$\eta_m = \eta_s e^{(\lambda \phi)}, \quad (\text{B.10})$$

with $\lambda = -29 \pm 3$ as an experimentally derived porosity-weakening coefficient (Katz et al., 2006). Eq. (B.10) is applicable for porous matrices when the fluid fraction is relatively low (<10%).

To resolve events ranging from slow (cm/year) to fast (m/s), we use adaptive time stepping (Dal Zilio et al., 2022a): the minimum of the slip acceleration on the fault (Δt_s), the displacement per grid cell (Δt_d) and the visco-elasto-plastic time step (Δt_{vep}). Additionally, it is suggested that both nucleation size and cohesive zone size are supposed to be well resolved (Lapusta et al., 2000). As most papers of foreshock sequences discussed (Cattania and Segall, 2021), the ratio of cohesive zone size to grid size should be $\Lambda_0/\Delta y > 3$. In this study, we use $\Lambda_0/\Delta y > 3.5$ for all models. For a comprehensive explanation of the methodology, including terms, governing equations, rheology, and the time-stepping algorithm, we refer to Dal Zilio et al. (2022a).

Appendix C. Supplementary simulation results

C.1. Slip rate evolution

In the main model, we showed slip-rate evolution as a function of time to the mainshock or swarm burst to emphasize the rupture process. Here, to illustrate the slip pattern throughout the entire seismic cycle, we show slip-rate evolution as a function of timestep for models M4–M6 (Fig. C.1). The timestep 10×10^4 in Fig. C.1b and c corresponds to 20 years in actual time.

C.2. Seismic ratio

In nature, l_k and t_k are poorly constrained due to the complex processes that increase or decrease permeability and the difficulty of direct measurement. Estimates from hydraulic level changes during observed earthquakes suggest that permeability recovery to pre-seismic levels can take from hours to years (Manga et al., 2012). Little evidence exists regarding the range of l_k . In this study, to examine the influence of l_k and t_k on fault slip behavior over a broad range, we test t_k from 10 h to 10 years and l_k from 0.1 m to 1 m (Zhu et al., 2020). Fig. C.2 shows that the seismic ratio inversely correlates with t_k , whereas no clear relationship is observed with l_k .

C.3. Scaling relationship

Apart from Fig. 7, we find that seismic moment scales with rupture length and duration. Fig. C.3a shows the relationship $L \propto M_0^{1/3}$, consistent with Denolle and Shearer (2016). Fig. C.3b shows that duration scales with seismic moment as $T_d \propto M_0^{1/2}$, falling between the ranges of regular and slow earthquakes reported by Ide et al. (2007). This indicates that the interaction of seismic and aseismic slip can bridge the gap between typical fast and slow events. Accordingly, the permeability heterogeneities included in our study likely provide

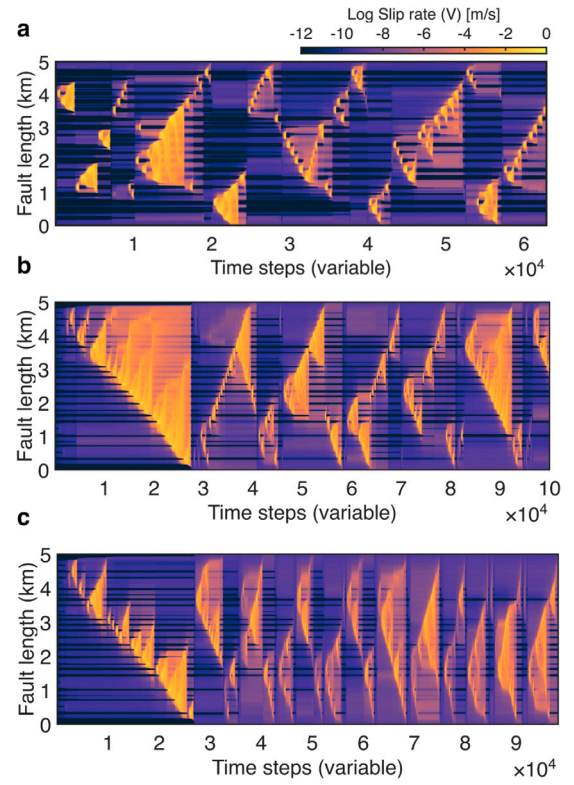


Fig. C.1. Slip rate evolution. Slip rate in log scale on the fault in time steps for Model M4 (a), Model M5 (b) and Model M6 (c), respectively.

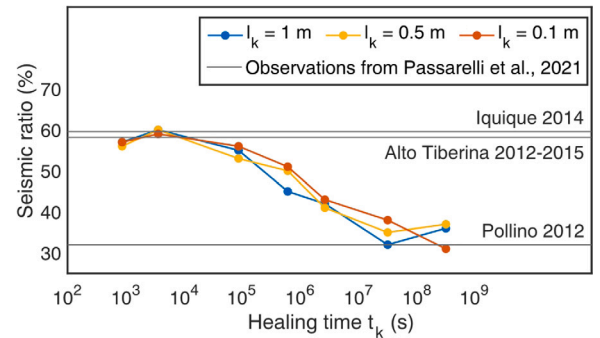


Fig. C.2. Seismic ratio of different healing time t_k and permeability enhancement evolution distance l_k , compared to observations. Seismic ratio decreases with the increase in t_k from 10 h to 10 years, while the ratio does not affect by the l_k from 0.1–1 m. The horizontal lines represent the seismic ratio (productivity) observed in various tectonic settings (Passarelli et al., 2021).

a plausible explanation for more complex slip behaviors observed in natural earthquake sequences.

Data availability

Data will be made available on request.

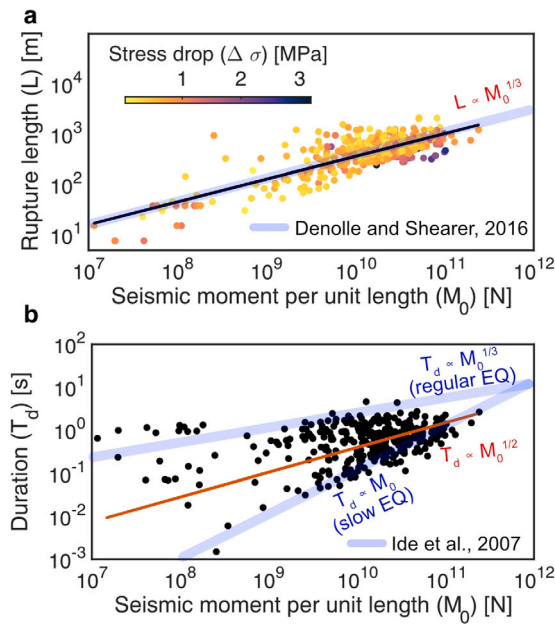


Fig. C.3. Comparison of scaling properties between the numerical experiments and observations. (a) Seismic moment M_0 versus rupture length L , with both simulated seismicity (dots and black line) and observed shallow thrust earthquakes (Denolle and Shearer, 2016) (blue line) resulting in $L \propto M_0^{1/3}$. The color map shows the stress drop of all events. (b) Moment-duration scaling of simulation (black dots and red line) lies between observation of slow earthquakes and regular earthquakes (Ide et al., 2007) (blue lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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